

FitBuddy – AI Fitness Plan Generator using Gemini Models

6. Frontend Development & Jinja2 Templates

User Information:

Description: Once the user submits their details, the application navigates to a personalized workout plan page. This section displays a summary of the user's input including their Name, User ID, Age, Weight, Fitness Goal, and Selected Intensity. This personalized display ensures users feel acknowledged and guides the AI engine to generate relevant workout recommendations tailored specifically to their fitness journey.

Workout Plan:

Description: FitBuddy generates a personalized, day-wise workout plan tailored to the user's fitness goal, experience level, and preferred workout schedule. The workout section on the result page outlines each day with a clear focus, such as full body, upper or lower body, cardio, core strength, or flexibility. For every workout day, a curated list of exercises is provided, detailing the name of the exercise, the number of sets and repetitions (or duration), and suggested rest intervals. The plan incorporates variety and structured progression to keep the workouts engaging and effective. Additionally, basic warm-up and cool-down guidance may be included to support safe and well-rounded training. Overall, this section aims to deliver a goal-oriented, easy-to-follow workout schedule that promotes consistency and progress.

Nutrition Tip:

Description: The Nutrition Tip section in FitBuddy provides users with concise, practical dietary advice to complement their workout plans. It emphasizes the importance of prioritizing protein intake in every meal, suggesting sources like chicken, fish, beans, or Greek yogurt. The tip explains how protein contributes to muscle building, satiety, and metabolic support—key factors for both fat loss and muscle gain. This section is designed to reinforce healthy eating habits that align with users' fitness goals.

Feedback Page:

Description: The Feedback Page in FitBuddy allows users to personalize and refine their fitness journey by submitting feedback on their current workout plans. Users are prompted to enter their unique User ID and provide specific feedback regarding the effectiveness, difficulty, or preferences related to their plan. Once submitted, this feedback is processed by the system to intelligently update and adjust the workout plan to better align with the user's goals, comfort, and progress. This dynamic feedback loop ensures that each fitness plan remains adaptive and tailored to the individual's evolving needs.

Once the plan is updated it comes with a confirmation message .

View all users page:

Description: The View All Users page in FitBuddy provides an overview of the entire user database, displaying detailed information for each registered user. This includes their personal details such as name, age, and fitness goals, along with their original workout plan generated at registration and the updated workout plan if any changes were made based on their submitted feedback. This feature enables a transparent view of how each user's plan has evolved over time, showcasing the adaptability of the FitBuddy system and helping administrators or developers monitor user progress and engagement effectively.

Conclusion:

The FitBuddy – AI Fitness Plan Generator is a comprehensive, user-centric web application that showcases the powerful synergy between artificial intelligence and health-tech solutions. Designed to assist individuals in achieving their fitness goals, FitBuddy uses Google's Gemini 1.5 Pro and Gemini Flash models to generate personalized 7-day workout plans and daily nutrition or recovery tips based on user inputs such as age, weight, fitness goal, and preferred workout intensity.

The application is built using the FastAPI framework for backend logic and API handling, SQLAlchemy with SQLite for persistent data storage, and Jinja2 templating for a clean and responsive frontend. Upon submitting their details on the homepage, users receive a tailored workout plan that includes warm-ups, main workouts, cooldowns, and goal-aligned tips. What sets FitBuddy apart is its dynamic feedback feature—users can provide input on the effectiveness or difficulty of their current plan, and the system responds by intelligently updating the plan using generative AI, ensuring ongoing customization and engagement.

Additionally, the application includes an administrative view where all registered users and their respective plans (original and updated) are displayed in a structured format. This functionality supports transparency, monitoring, and further optimization, making FitBuddy suitable for both individual users and coaches or institutions managing multiple fitness profiles.

Overall, FitBuddy stands as a scalable and adaptable AI-powered solution that not only personalizes the fitness journey but also evolves with the user's progress and preferences. It demonstrates how cutting-edge technology can be integrated into everyday wellness routines to promote healthy, consistent, and goal-oriented lifestyles.